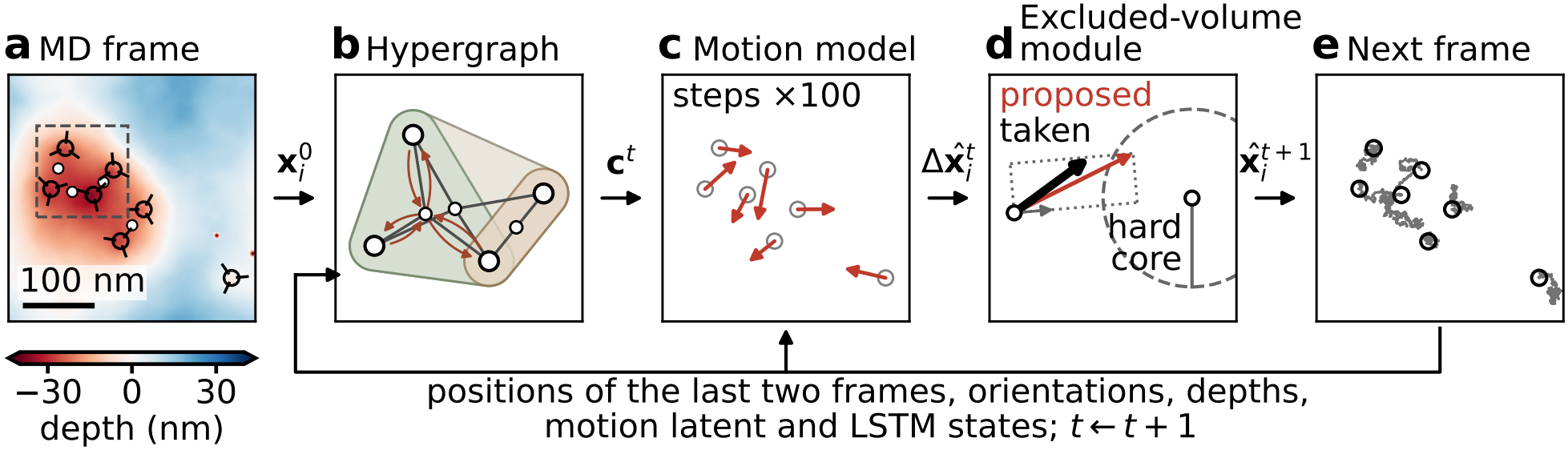

a MD frame
100 nm
−30 0 30
depth (nm)
$\mathbf{x}_i^0$
b Hypergraph
$\mathbf{c}^t$
c Motion model
steps ×100
$\Delta\hat{\mathbf{x}}_i^t$
d Excluded-volume module
proposed
taken
hard core
$\hat{\mathbf{x}}_i^{t+1}$
e Next frame
positions of the last two frames, orientations, depths,
motion latent and LSTM states; $t \leftarrow t+1$